

OTA-C based

The Second Order low pass filter

its transfer function is:

$$H_{lp}(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

s = Laplace variable

ω_0 = Angular frequency

Q = Quality factor

OTA equation:-

ideal OTA produces output current:

$$i_o = g_m (V_+ - V_-)$$

g_m = transconductance

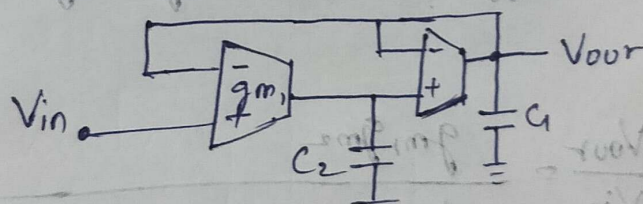
Capacitor equation:-

$$i_c = C \frac{dv}{dt}$$

In Laplace

$$i_c = sCV$$

OTA + capacitor can behave like integrator



The equation of C_2 :

$$i_2 = g_{m2} (V_{in} - V_{out})$$

$$i_{C2} = sC_2 V_x$$

$$\therefore sC_2 V_x = g_{m2} (V_{in} - V_{out}) \rightarrow (1)$$

The equation of C_1 :

$$i_1 = g_{m1} (V_x - V_{out})$$

$$sC_1 V_{out} = g_{m1} (V_x - V_{out})$$

$$sC_1 V_{out} = g_{m1} (V_x - V_{out}) \rightarrow (2)$$

$$sC_1 V_{out} = g_{m1} V_x - g_{m1} V_{out}$$

$$sC_1 V_{out} + g_{m1} V_{out} = g_{m1} V_x$$

$$V_{out} (sC_1 + g_{m1}) = g_{m1} V_x$$

$$V_x = \left(\frac{sC_1 + g_{m1}}{g_{m1}} \right) V_{out} \rightarrow (3)$$

Substitute @ 3 in 1?

$$sC_2 V_x = g_{m2} (V_{in} - V_{out})$$

$$sC_2 \left(\frac{sC_1 + g_{m1}}{g_{m1}} \right) V_{out} = g_{m2} (V_{in} - V_{out})$$

$$sC_2 (sC_1 + g_{m1}) V_{out} = g_{m1} g_{m2} (V_{in} - V_{out})$$

$$(s^2 C_1 C_2 + sC_2 g_{m1}) V_{out} = g_{m1} g_{m2} V_{in} - g_{m1} g_{m2} V_{out}$$

$$V_{out} (s^2 C_1 C_2 + sC_2 g_{m1} + g_{m1} g_{m2}) = g_{m1} g_{m2} V_{in}$$

$$\frac{V_{out}}{V_{in}} = \frac{g_{m1} g_{m2}}{s^2 C_1 C_2 + sC_2 g_{m1} + g_{m1} g_{m2}}$$

Divide with $C_1 C_2$ with everything

$$\frac{V_{out}}{V_{in}} = \frac{g_{m1} g_{m2}}{s^2 + s \frac{g_{m1}}{C_1} + \frac{g_{m1} g_{m2}}{C_1 C_2}}$$

$$(100V - 5V) \text{ imp} = 1$$

$$\therefore H(s) = \frac{g_{m1} g_{m2}}{C_1 C_2} \frac{1}{s^2 + \frac{g_{m1}}{C_1} s + \frac{g_{m1} g_{m2}}{C_1 C_2}} \quad (4)$$

Compare (4) with second order equation

$$\omega_0^2 = \frac{g_{m1} g_{m2}}{C_1 C_2}$$

$$\omega_0 = \sqrt{\frac{g_{m1} g_{m2}}{C_1 C_2}}$$

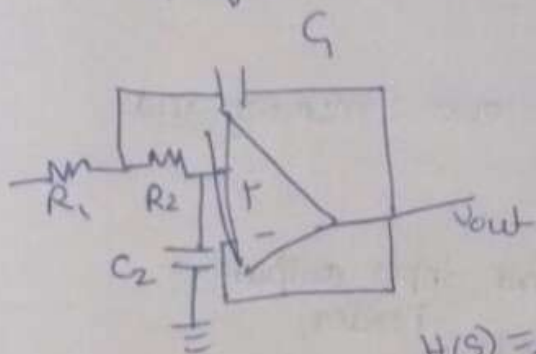
$$\frac{\omega_0}{Q} = \frac{g_{m1}}{C_1}$$

$$Q = \frac{\omega_0 C_1}{g_{m1}}$$

$$Q = \frac{C_1}{g_{m1}} \sqrt{\frac{g_{m1} g_{m2}}{C_1 C_2}}$$

$$Q = \sqrt{\frac{g_{m2} C_1}{g_{m1} C_2}}$$

op amp base calculations
 & equation



$$H(s) = \frac{1}{1 + sC}$$

$$T = RC$$

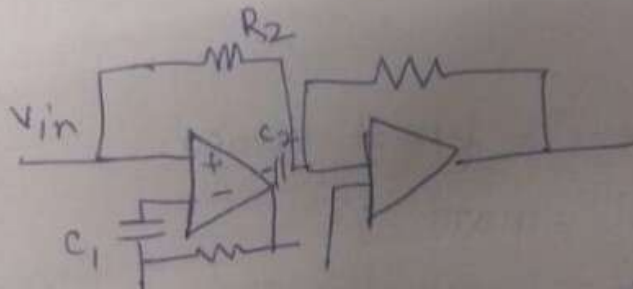
$$\omega_c = \frac{1}{RC}$$

$$f_c = \frac{1}{2\pi RC}$$

first order

$$H(s) = \frac{sRC}{1 + sRC}$$

$$f_c = \frac{1}{2\pi RC}$$



$$Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{(R_1 + R_2) C_2}$$

$$Q = \frac{\omega_0}{BW}$$

$$H(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$f_0 = \frac{\omega_0}{2\pi}$$

$$f_0 = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$

LP =

$$H_{LP} = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

2nd inverse

$$H_{inv}(s) = \frac{s^2 + BWs + \omega_0^2}{s^2 + QBWs + \omega_0^2}$$

$H_p(s)$

$$= \frac{s^2 + QBWs + \omega_0^2}{s^2 + BWs + \omega_0^2}$$

CFOA - Based Second-Order Peaking Filter Derivation:

time t

* Ideal CFOA Equations:

$$V_x = V_y$$

$$I_y = 0$$

$$I_z = I_x$$

output voltage $V_o = V_z$

* Required second order peaking filter equation

$$H_p(s) = \frac{s^2 + G_1 BW s + \omega_0^2}{s^2 + BW s + \omega_0^2}$$

$\Rightarrow \omega_0 = \text{peak angular freq.}$

$BW = \text{bandwidth}$

$G_1 = \text{maximum gain}$

$$H_{inv,p}(s) = \frac{s^2 + BW s + \omega_0^2}{s^2 + G_1 BW s + \omega_0^2}$$

* Transfer function:

$$H(s) = \frac{s^2 + \frac{1}{R_3 C_1} s + \frac{1}{R_2 R_3 C_1 C_2}}{s^2 + \frac{1}{R_1 C_1} s + \frac{1}{R_2 R_3 C_1 C_2}}$$

* Complete denominator terms:

Standard: $s^2 + BW s + \omega_0^2$

CFOA: $s^2 + \frac{1}{R_1 C_1} s + \frac{1}{R_2 R_3 C_1 C_2}$

Hence $\Rightarrow$ $BW = \frac{1}{R_1 C_1}$

$$\omega_0^2 = \frac{1}{R_2 R_3 C_1 C_2}$$

$$\omega_0 = \frac{1}{\sqrt{R_2 R_3 C_1 C_2}}$$

* Numerator:

standard: $s^2 + G BW s + \omega_0^2$

CFOA: $s^2 + \frac{1}{R_3 C_1} s + \frac{1}{R_2 R_3 C_1 C_2}$

$G BW = \frac{1}{R_3 C_1}$ & $BW = \frac{1}{R_1 C_1}$

$$\omega_0 = \frac{1}{\sqrt{R_2 R_3 C_1 C_2}}$$

$$G \left(\frac{1}{R_1 C_1} \right) = \frac{1}{R_3 C_1}$$

$$\frac{G}{R_1} = \frac{1}{R_3}$$

$$G = \frac{R_1}{R_3}$$

* for CFOA resonator

$$H_p(s) = \frac{s^2 + \frac{1}{R_3 C_1} s + \frac{1}{R_2 R_3 C_1 C_2}}{s^2 + \frac{1}{R_1 C_1} s + \frac{1}{R_2 R_3 C_1 C_2}}$$

$$\Rightarrow \boxed{\omega_0 = \frac{1}{\sqrt{R_2 R_3 C_1 C_2}}} \quad \boxed{BW = \frac{1}{R_3 C_1}} \quad \boxed{G = \frac{R_1}{R_3}}$$

* Resonant frequency (f_0):

wkt, $\omega_0 = 2\pi f_0$

$$f_0 = \frac{\omega_0}{2\pi}$$

$$\Rightarrow \boxed{f_0 = \frac{1}{2\pi \sqrt{R_2 R_3 C_1 C_2}}}$$

ω_0 angular freq.

$$\omega_0 = \frac{1}{\sqrt{R_2 R_3 C_1 C_2}}$$

* Quality factor (Q):

$$Q = \frac{\omega_0}{BW}$$

$$\omega_0 = \frac{1}{\sqrt{R_2 R_3 C_1 C_2}}$$

$$BW = \frac{1}{R_1 C_1}$$

$$Q = \frac{\frac{1}{\sqrt{R_2 R_3 C_1 C_2}}}{\frac{1}{R_1 C_1}}$$

$$\Rightarrow \boxed{Q = \frac{R_1 C_1}{\sqrt{R_2 R_3 C_1 C_2}}}$$

* Anti Resonator:

$$H_{inv,p}(s) = \frac{s^2 + BW + \omega_0^2}{s^2 + G BW s + \omega_0^2}$$

By compalision:

$$G BW = \frac{1}{R_1 C_1}$$

$$BW = \frac{1}{R_3 C_1}$$

$$\Rightarrow G \left(\frac{1}{R_3 C_1} \right) = \frac{1}{R_1 C_1}$$

$$\Rightarrow G = \frac{R_3}{R_1}$$

and center frequency remains,

$$\omega_0 = \frac{1}{\sqrt{R_2 R_3 C_1 C_2}}$$

$$\frac{R_3}{R_1} = \beta$$